

PROGRAM 5

AIM:-K-NN CLASSIFICATION

Question 1:- Program to implement k-NN classification using any standard dataset available in the public domain and find the accuracy of the algorithm.

Code1

```
import numpy as np

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# df=pd.read_csv('data.csv')

data = load_iris()

X = data.data

y= data.target

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)

# k=3

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train,y_train)

y_pred = knn.predict(X_test)

accuracy = accuracy_score(y_test,y_pred)

print(f'Accuracy of k-NN with k={k}:{accuracy:.2f}')
```

not for record for reference

```
#Load the breast cancer dataset
```

```
import numpy as np  
  
from sklearn.datasets import load_breast_cancer  
from sklearn.model_selection import train_test_split  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.metrics import accuracy_score  
from sklearn.preprocessing import LabelEncoder  
from sklearn.preprocessing import LabelEncoder
```

```
#Load the breast cancer dataset
```

```
data=load_breast_cancer()  
  
X=data.data #Features  
y=data.target #Target(labels)
```

```
#Split the dataset into training and testing sets
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

```
#Initialize the k-NN classifier with a specified value of k
```

```
k=3 #You can change this value as needed  
  
knn=KNeighborsClassifier(n_neighbors=k)
```

```
#Fit the model on the training data
```

```
knn.fit(X_train,y_train)
```

```
#Make predictions on the test data
```

```
y_pred=knn.predict(X_test)
```

```
#Calculate the accuracy of the model
```

```
accuracy=accuracy_score(y_test,y_pred)
```

```
print(f'Accuracy of k-NN with k={k}:{accuracy:.2f}')
```

```
#load foo csv
```

```
import numpy as np
```

```
import pandas as pd
```

```
# from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = pd.read_csv('food.csv')
```

```
X = data.iloc[:, :3]
```

```
# print(X)
```

```
y= data.iloc[:, 3]
```

```
le = LabelEncoder()
```

```
categorical_columns = ['Ingredient']
```

```
for col in categorical_columns:
```

```
    X[col] = le.fit_transform(X[col])
```

```
y = le.fit_transform(y)
```

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
```

```
k=3
```

```
knn = KNeighborsClassifier(n_neighbors=k)
```

```
knn.fit(X_train,y_train)
```

```
y_pred = knn.predict(X_test)
```

```
sample=[[1,10,9]]
```

```
k=knn.predict(sample)
```

```
print(k)
```

```
accuracy = accuracy_score(y_test,y_pred)
```

```
print(accuracy)
```

```
#load insurance
```

```
import numpy as np
```

```
import pandas as pd
```

```
# from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
data = pd.read_csv('insurance.csv')
```

```
X = data.iloc[:, :6]
```

```
# print(X)
```

```
y= data.iloc[:, 1]
```

```
le = LabelEncoder()
```

```
categorical_columns = ['sex', 'smoker', 'region']
```

```
for col in categorical_columns:
```

```
    X[col] = le.fit_transform(X[col])
```

```
y = le.fit_transform(y)
```

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
```

```
k=3
```

```
knn = KNeighborsClassifier(n_neighbors=k)
```

```
knn.fit(X_train,y_train)
```

```
y_pred = knn.predict(X_test)
```

```
sample=[[1,10,9,11,4,2]]
```

```
k=knn.predict(sample)
```

```
print(k)
```

```
accuracy = accuracy_score(y_test,y_pred)
```

```
print(accuracy)
```

```
#cricket
```

```
import numpy as np

import pandas as pd

# from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

from sklearn.preprocessing import LabelEncoder


data = pd.read_csv('cricket.csv')

X = data.iloc[:, :5]

# print(X)

y = data.iloc[:, 4]

# print(y)


le = LabelEncoder()

categorical_columns = ['Outlook', 'Temp', 'Humidity', 'Windy', 'Play Cricket']

for col in categorical_columns:

    X[col] = le.fit_transform(X[col])

y = le.fit_transform(y)


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

k=3

knn = KNeighborsClassifier(n_neighbors=k)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

sample=[[1,10,9,11,4]]
```

```
k=knn.predict(sample)
print(k)
accuracy = accuracy_score(y_test,y_pred)
print(accuracy)
```